

- 1(a) A charity uses a desktop computer to record financial donations that it receives. The computer contains a single core, 2.4GHz processor with 2MB cache.

A processor uses the Von Neumann architecture.

- i. Describe what is meant by the term 'Von Neumann architecture'.

----- [2]

- ii. Give **one** way that the Harvard architecture differs from the Von Neumann architecture.

----- [1]

- (b) A charity is concerned that the performance of a computer is not sufficient and wishes to replace the processor.

Give **two** features of a replacement processor that would increase the typical performance of the computer.

1 -----

2 -----

[2]

- (c) **Fig. 1** shows assembly code written using the Little Man Computer (LMC). The program calculates and outputs the total amount that is donated to a charity in any particular day.

Depending on the amount, an additional bonus may be added to each amount donated.

start	INP
	STA donation
	SUB hundred
	BRP bonus
nobonus	LDA total
	ADD donation
	STA total
	OUT
	BRA start
bonus	LDA total
	ADD donation
	ADD twenty
	STA total
	OUT
	BRA start
hundred	DAT 100
twenty	DAT 20
donation	DAT 0
total	DAT 0

Fig. 1

- i. The program shown in **Fig. 1** is run **once** using **three** different inputs. Therefore, while the program is running once, it will output the updated total three times.

Give the total values that are output when the values **10**, **50** and **120** are input into this program.

Output for 10 -----

Output for 50 -----

Output for 120 -----

[3]

- ii. Write LMC code that will reset the value of the memory location labelled **total** to zero and then stop the program.

[4]

iii. This program is run on a processor that allows pipelining.

Define the term 'pipelining'.

[3]

iv. Explain **one** benefit to a charity of using a processor that allows pipelining.

[2]

(d) The processor contains registers including the accumulator and the program counter. The contents of these registers are modified during the Fetch-Decode-Execute cycle.

i. Describe how the accumulator is used during the Fetch-Decode-Execute cycle.

[2]

ii. Describe how the program counter is used during the Fetch-Decode-Execute cycle.

[2]

iii. State the name of **three** other registers that are used during the Fetch-Decode-Execute cycle.

1 -----

2 -----

3 -----

[3]

- (e) * A charity has several desktop computers in their office that use a CISC processor. They are considering buying mobile devices for their staff to use when they are not in the office.

Discuss whether these mobile devices should use the same CISC processors that are used in their desktop computers or if they should use a RISC processor instead.

You should include the following in your answer:

- the difference between each processor type
- the suitability of each processor type for mobile devices.

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

This image shows a full page of a document template. It consists of approximately 28 horizontal dashed lines spaced evenly down the page, providing a guide for handwriting or typing. The lines are thin and black against a plain white background.

2(a) A video streaming service uses a relational database. An extract of the data from two tables from this database is shown in Fig. 2.

Membership contains data about current memberships that customers hold and package contains data about different streaming packages available.

Username	FirstName	StartDate	PackageType
User001	Amaya	08/05/2016	Premium
User002	Amit	06/06/2019	Basic
User003	Tom	17/08/2019	Free
User004	Kareem	08/08/2017	Basic
User005	Sarah	25/03/2020	Premium

Membership

PackageType	CostPerMonth (£)	Adverts
Premium	12.99	false
Basic	7.99	true
Free	0.00	true

Package

Fig. 2

i. State what is meant by the term ‘primary key’.

[1]

ii. Identify the foreign key used in the database and the table name where this is a foreign key.

Foreign Key -----

Table Name -----

[2]

iii. Identify the data type of the `CostPerMonth (£)` field.

----- [1]

iv. Give the name of the field that could be stored using a Boolean data type.

----- [1]

- (b) The `Adverts` field indicates if customers will be shown adverts. `true` indicates that customers will be shown adverts, and `false` indicates that adverts are not shown.

Write Structured Query Language (SQL) to return the `Username` and `FirstName` fields for all customers who see adverts.

----- [5]

(c) When new customers join the streaming service, their name, email address and contact details are captured so that they can be entered into the database.

i. Identify **one** method of capturing a new customer’s personal data, describing why this method is suitable.

Method -----

Suitability -----

[3]

ii. Sometimes the company may need to move or backup its data they hold about customers.

Identify **two** methods of exchanging data with other computer systems.

1 -----

2 -----

[2]

(d) A relational database supports ACID transactions. ACID stands for Atomicity, Consistency, Isolation and Durability.

i. Describe what is meant by a transaction being durable.

[2]

ii. Give **one** way that durability can be achieved for a completed transaction.

[1]

iii. Explain how record locking can be used to ensure that the ACID principle of isolation is achieved when carrying out multiple transactions.

[3]

iv. Give **one** disadvantage of using record locking.

[1]

(e) The Copyright Designs and Patents Act 1988 applies to all videos that are streamed.

Explain how this act applies to the videos.

[2]

(f) All videos that are streamed are compressed. Customers have the option to choose from watching the videos with lossy compression or lossless compression.

Explain how this choice will impact the customer.

[5]

- (g) A program is written using an object-oriented programming paradigm and uses a class called `video` to organise videos that are streamed to customers.

The class `video` has these attributes:

- name
- number of views
- star rating.

The constructor method will set the name attribute to the name that is passed in as a parameter. The constructor will also initially set the number of views to 0 and the star rating to 3.

- i. Write program code or pseudocode to declare the class `video` and initialise the required attributes as private.

You should include **both** the attribute definitions and the constructor method in your answer.

[7]

- ii. A public method called `updateviews()` will update the number of views after a video has been viewed.

This method is defined inside the `video` class.

Write program code or pseudocode for the method `updateviews()` to increase the number of views by one.

[2]

3(a)

- i. Convert the hexadecimal value **B7E** to a binary number.

----- [1]

- ii. 110010101 is a binary number that is represented using sign and magnitude.

Convert this binary number to a denary number.

----- [1]

- iii. Complete this binary subtraction. Both numbers are 8-bit integer values represented using two's complement.

Show the result in the same format and show your working.

0110 1101 —
0011 0100

----- [3]

- Give the denary version of this number, showing your working.

[4]

- Tick (✓) one box in each row to state whether each number is normalised or not normalised.

Binary number	Normalised	Not normalised
010101 100		
110101 111		
011010 010		
101010 110		

Table 3

[4]

4 * Amit is studying Computer Science at university. He has been asked to write an assignment on Artificial Intelligence (AI).

Discuss the extent to which you think computer systems will inherit the biases and discrimination of their programmers as the use of AI increases.

You should include the following in your answer:

- the meaning of AI
- examples of when AI may be affected by bias
- the measures that can be taken to prevent people being affected by bias in AI.

Blank lined paper for writing.

5(a) A programmer creates this function shown in Fig. 5 using a high-level language.

```
function mystery(x,y)

    total = x + y

    while x >= 10 then

        x = x - 10

        y = y - 10

        total = total + x + y

    endwhile

    return total

endfunction
```

Fig. 5

- i. State the value output by the line `print(mystery(10,20))`

----- [1]

- ii. State the value output by the line `print(mystery(0,70))`

----- [1]

- iii. State the value output by the line `print(mystery(45,55))`

----- [1]

(b) Before the code in Fig. 5 can be executed, a translator must be used.

i. State the purpose of a translator.

[1]

ii. Explain **two** differences between a compiler and an interpreter.

Difference 1 -----

Difference 2 -----

[4]

(c) For each statement shown in **Table 5**, tick (✓) **one** box in each row to indicate which stage of compilation each action takes place at.

	Lexical analysis	Syntax analysis	Code generation
Comments and whitespace are removed			
Keywords are replaced with tokens			
Object code is created			
Symbol table created for variables			
Builds an abstract syntax tree			

Table 5

[5]

(d) Describe the purpose of code optimisation.

[2]

- (e) A programmer creates another function to count and return how many capital letters are in a string that is passed into the function as a parameter.

The `asc ( )` function takes in a character and returns its ASCII value. For example `asc ( "A" )` returns 65. Capital letters have ASCII values between 65 and 90 inclusive.

- i. Complete the function below.

```
function countCapitals(text)
    // initialise counter to 0
    capCount = 0
    // loop through each character in the string passed in
    for x = 0 to text.length-1
        c = text.subString(x, 1)
        // check if character is a capital
        if asc(c) >= 65 .....
            // if so, increment counter
            .....
        endif
    next x
    .....
endfunction
```

[3]

- ii. Give **one** similarity between ASCII and Unicode.

----- [1]

- iii. Give **two** differences between ASCII and Unicode.

Difference 1 -----

Difference 2 -----

[2]

(f) * A programmer has been asked by a client to create a complex computer program. Compare the spiral model and waterfall lifecycle methodologies for this task.

You should include the following in your answer:

- how both methodologies could be used to develop a complex computer program
- the benefits of each methodology for this task
- the drawbacks of each methodology for this task.

Blank lined paper for writing.

6(a) Anika’s computer runs a multi-tasking operating system. She has access to a printer and a broadband internet connection through a wireless connection. The operating system uses scheduling algorithms such as first come first served and round-robin.

i. Explain why the computer’s operating system uses a first come first served algorithm when sending documents to the printer.

[2]

ii. Explain why the computer’s operating system uses a round-robin algorithm for allocating processor time.

[3]

iii. Describe **one** other scheduling algorithm.

[2]

(b) Anika uses an encrypted messaging program to communicate with her friends. The computer uses the TCP/IP stack.

i. Explain what happens at the application layer of the TCP/IP stack when using this program.

[5]

ii. Explain what happens at the link layer (sometimes referred to as the “network interface layer”, “network access layer” or simply the “network layer”) of the TCP/IP stack when using this program.

[2]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a	i	- Both <u>data and instructions</u> share the same memory <u>Instructions and Data</u> stored in same format - A single set of buses / same <u>bus for instructions & data</u> (to connect CPU to Memory and I/O) - Has a (single) control unit - Has an ALU. - Has ways to input and output. - Has access to storage, - Works sequentially through instructions // follows Fetch-execute cycle (Special) registers within CPU - Based on stored program concept	2	Examiner's Comments Many candidates were able to access full marks on this question. This question has been asked in previous papers and candidates should be encouraged to use these to make sure they are clear in their responses. There were many possible responses in the mark scheme to help candidates to gain full marks. Most candidates gained at least 1 mark.
		ii	- Separate <u>memory for data and instructions</u> / Multiple memory units - Different (sets of) buses one for <u>instructions</u> & one for <u>data/ instructions and data</u> can be accessed concurrently.	1	Examiner's Comments This question was generally answered well by candidates and the majority gave separate areas of memory for data and instructions. Where candidates were not given marks, it was generally because their answer was unclear, e.g. just saying 'separate memory'.
	b		Higher/faster clock speed More cores//dual/quad/etc core More cache memory.	2	Answers must refer to an improvement (more/higher/faster) not just "change the clock speed" Allow discussions of level 1/level 2 cache sizes for one mark. Accept valid features of CPUs that would improve performance e.g. Use of: Pipelining Simultaneous Multithreading Do not accept RISC/CISC. Examiner's Comments Most candidates were able to gain full marks on this question. Less successful responses often mentioned clock speed, cache or cores without referring to an improvement, e.g. higher or faster.
	c	i	10 60 200	3	1 mark per number Examiner's Comments This was generally well answered by candidates who had a good understanding of LMC. Candidates should be encouraged to trace through LMC programs with different values as well as writing them.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	• Loads a value into the accumulator • Establishes a zero value (by use of DAT / SUB) • Stores a <u>zero value</u> into total • Program stops	4	Example 1 <pre>LDA zero STA total HLT zero DAT 0</pre> Example 2 <pre>LDA total SUB total STA total HLT</pre> BP1 can be given for any value being loaded into the accumulator e.g. INP If candidate writes LDA donation/total (case sensitive) they can get BP2 as they've used the labels from the question BP3 - total is case sensitive as given in the question BP4 - must not be given if the zero value will be attempted to be fetched e.g. HLT is placed after DAT <u>Examiner's Comments</u> This was generally answered well, and the majority of students were able to gain marks with most gaining 3 or 4 marks. Less successful responses over complicated the program leading to them making mistakes. A small number of candidates attempted to answer in pseudocode rather than LMC. Candidates should be encouraged to use the commands in Appendix 5d of the specification.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	- One instruction can be fetched while another is being decoded... ...and another is executed - The output of one <u>process/instruction</u> is the input of the next. - Concurrent processing of multiple instructions // completing multiple FDE cycles at once	3	For BP1, allow any 2 of the 3 parts of the FDE cycle For BP2, must give the other part of the FDE cycle not given in BP1 Do not award if explaining multiple cores working on different parts of FDE cycle Examiner's Comments Many candidates were able to gain at least 2 marks on this question. Some candidates were not awarded marks as they wrote about multiple cores or programs being fetched instead, of instructions. Exemplar 1 Pipelining is when a computer can fetch the next instruction whilst the previous is being decoded and the one before that is being executed. There is two types of pipelining: arithmetic and instruction. Pipelining allows multiple instructions to be processed at the same time. [3] The candidate has clearly described pipelining with correct terminology. They gained full marks for the description of one <u>instruction</u> being decoded while another is fetched and another is executed, as well as describing that it allows <u>multiple instructions</u> to be processed <u>at the same time</u>.
		iv	- More <u>instructions</u> can be carried out in a set amount of time // less time to execute the same number of <u>instructions</u> - Increasing the speed/performance/efficiency of the computer/program // quicker for the program to complete	2	Do not allow "each instruction is quicker to execute". BP2 has to be specific to the charity e.g. processing more donations Examiner's Comments Many candidates gained 1 mark for giving a benefit to the charity, but they did not go on to say why pipelining enabled that. Some candidates did not apply their answer to the charity, so were not awarded the mark for the benefit.
	d	i	- Holds all input/output - Holds results of calculations (from the ALU) - Checked for conditional branching (e.g. BRZ) - Stores data which has come from the MDR/RAM	2	Examiner's Comments Most candidates were able to access 1 mark for the result of ALU calculations, but few were able to give two uses. Some confused the accumulator with the program counter and the ALU.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	• Holds the <u>address/location</u> of the <u>next</u> instruction (to be executed/fetched) • Contents copied to the MAR at start of FDE • Incremented (by one) on every cycle • Can be changed by branch/jump instructions	2	Examiner's Comments This question was generally well answered by candidates who gave clear responses. Misconception  Some candidates thought that the program counter kept track of a count of the number of instructions that had been fetched.
		iii	• Memory Address Register // MAR • Memory Data Register // MDR • Current Instruction Register // CIR • Index Register // IR	3	Allow Memory Buffer Register for MDR Examiner's Comments Most candidates gained full marks on this question and were able to correctly identify three other registers. Some lost marks for saying the ALU or control unit were registers.
	e		Mark Band 3–High Level (9-12 marks) The candidate demonstrates a thorough knowledge and understanding of both CISC and RISC. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. Mark Band 2-Mid Level (5-8 marks) The candidate demonstrates reasonable knowledge and understanding of CISC and/or RISC; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation.	12 (AO1.1) (2), (AO1.2) (2), (AO2.1) (3), (AO3.3). (5)	AO1 CISC is a complex instruction set. The traditional approach to processor design. Lots of instructions available although some instructions in CISC will rarely get used. RISC is a reduced instruction set. A smaller number of instructions available, several instructions can be combined to perform the same tasks as CISC processors. RISC instructions are used regularly. RISC has fewer transistors/less complex circuitry whereas CISC integrated circuits are more expensive/complicated. RISC instructions take one cycle whereas CISC may take several. RISC can only do complex things by combining multiple instructions whereas CISC is done in one line. Compilers for RISC need to be more complex than compilers for CISC AO2 CISC processors would run the same software as the desktop machines. Would be less power efficient and require larger battery and cooling mechanisms. More expensive to purchase. RISC processor requires software to be written specifically for it (cannot use CISC instructions). More power efficient and so requires less/no cooling and smaller battery/longer battery life. RISC devices may require greater RAM as programs tend to be larger than their CISC

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. Mark Band 1-Low Level (1-4 marks) The candidate demonstrates a basic knowledge of CISC or RISC; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 marks No attempt to answer the question or response is not worthy of credit.		equivalents. AO3 Mobile use of CISC would save money on software and increase compatibility but cost more to purchase and be physically larger (heat sink/larger battery) and/or have a shorter battery life. RISC would require investment in software but be cheaper to purchase and give a better performance out of the office (lighter/longer battery life). Some compatibility issues may be reduced with emulators and translators. Examiner's Comments Many candidates were able to discuss the difference in reduced or complex instruction sets and gave some discussion of the increase in hardware requirements for CISC. Few talked about the software differences, and some assumed the charity would need to be programming the devices which was not relevant to the question.
			Total	36	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2	a	i	Field that is unique/does not repeat	1	
		ii	Foreign Key: PackageType Table Name: Membership	2	Must be spelled correctly <u>Examiner's Comments</u> Many candidates gained 1 mark for the foreign key and most gained both marks, although some candidates gave 'package' as the table where it is a primary key rather than the membership table where it is the foreign key.
		iii	Float / Floating Point / Real	1	Allow currency/double/single/decimal
		iv	Adverts	1	CAO

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	• Username and FirstName fields (and no others) selected correctly using SELECT keyword • Membership / both tables correctly selected using FROM keyword • Tables joined using correct JOIN / INNER JOIN keywords // Tables joined using correct WHERE clause • Fields use table identifiers before them • WHERE clause used to correctly show only records where Adverts = true	5	For full marks, a fully correct working answer must be provided. Candidates can join tables in either of two valid ways (using JOIN or WHERE). Note that JOIN is given in the specification but INNER JOIN is also equally acceptable. BP1 is the same for either method For BP2, candidates can either choose just the Membership table or <u>both</u> the Membership and Package table BP3 credited for correct JOIN / INNER JOIN or correct use of WHERE clause to join tables. <u>Do not credit if FROM clause incorrect for this method</u> BP4 credited if candidates have used table identifiers before the field name (i.e they have used Membership.PackageType and not just PackageType) BP5 will require use of AND if WHERE is used to join tables. Spellings of all field names, table names and keywords must be accurate but only penalise once. <u>Example one using JOIN keyword</u> SELECT Username, Firstname FROM Membership JOIN Package on Membership.PackageType=Package. PackageType WHERE Adverts = true <u>Example two using WHERE clause</u> SELECT Username, Firstname FROM Membership, Package WHERE Membership.PackageType = Package.PackageType AND Adverts = true <u>Examiner's Comments</u> Many candidates were able to gain some marks. The question refers to the Adverts field which is in the package table and states that the data shown in the tables is only an extract from the tables. For full marks on this question, candidates were expected to attempt to join the two tables to access the Username and Firstname from the membership table, and the Adverts from the package table.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c	i	- Form / web form - Can use validation to check for common errors... - Can check for duplicate values - Data can be entered direct into the database / limited manual processing - Can be done from remote locations	3	One mark maximum for identification of method Two marks for discussion of suitability Accept other valid methods of capturing data. Do NOT accept OCR/OMR/barcodes/QR codes If the method is incorrect, don't read on Examiner's Comments Few candidates were able to gain full marks on this question as many overcomplicated it or could not give a relevant data capture method. When they did give form as a relevant answer, they often had unclear suitability. Exemplar 2 Method <i>Electronic form filled out by customer</i> Suitability <i>An electronic form can be processed automatically and details new customer details automatically added to the database. An electronic form is also suitable for the customer as it can be filled in at home.</i> [3] The candidate has given a valid method and has given clear and correct suitability by describing that the details could be automatically added to the database and can be filled in from home, which would be a remote location. The candidate gained the full 3 marks.
		ii	e.g. - CSV - JSON - XML - SQL - APIs - EDI - RSS - SOAP	2	Examiner's Comments Few candidates gained full marks on this question although there were a range of relevant responses they could have given.
	d	i	(Committed) data/transaction is not lost... ...in case of power / system failure	2	
		ii	Completed transactions stored in secondary storage // data not stored long-term in RAM/cache	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	- The outcome of concurrent transactions is the same as if transactions were completed sequentially. - Record locking allows one user/process to access/modify record level data at any one time - So data that is being used elsewhere cannot be modified // data that is being modified elsewhere cannot be used	3	Allow reference to lost updates/dirty reads/phantom reads for BP3. Examiner's Comments For candidates with a good understanding of ACID, these questions were well answered. Unfortunately, some had only a vague knowledge or confused it with referential integrity. Some answers were unclear. Some candidates talked about locking the entire database when record locking rather than just the relevant records.
		iv	- Can cause delays (as users wait for access) - Can cause deadlock	1	Examiner's Comments Many candidates were given a mark for deadlock or longer wait times. Those candidates given a mark in Question 2 (d) (iv) tended to be those who has gained marks in Question 2 (d) (iii).
	e		- Copyright assigned to owner of video automatically on creation - Makes it illegal to copy/distribute videos as your own/without permission - Copyright holder can ask for their work to be removed from the streaming platform - Membership/licence gives subscribers the agreement to view videos - Which may restrict their use (e.g. to whom it is shown or geographical location from which it is accessed).	2	Examiner's Comments Most candidates were able to gain 1 mark for this question, but few went on to gain a second mark.
	f		- Lossy permanently removes data - Lossless rewrites original data in more efficient format - Lossless is able to recreate the original file // Lossy is not able to recreate the original file - Lossy reduces quality of videos // Lossless keeps original quality - Lossy file size is smaller than if lossless were used - Lossy: compression ratio may be adjusted depending on bandwidth - Resulting in a noticeable decrease in quality on slower connections. - Lossy: the video will buffer less / quicker to start watching the video // Lossless: the video will buffer more / slower to start watching the video	5	Do not allow answers relating to speed of download unless this clearly refers to the video starting or reduction in buffering – scenario is video being streamed, not downloaded. Examiner's Comments Candidates tended to write at length for this question, but often made the same point twice. Many missed marks for not making the comparison between lossy and lossless and only gave one side. Some candidates discussed the videos being downloaded rather than streamed.
	g	i	- Class definition with identifier <u>video</u> - name, number of views and star rating	7	Accept implementations in high-level languages (e.g. __ for private, class name

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
	attributes defined... • ...As private • Constructor method definition <u>inside class definition</u>... • ...that accepts only one parameter • ...Name attribute set to parameter passed in • Views set to 0 and rating set to 3 either when initialised or in constructor.		used for constructor, no need for end of class definition in Python) BP1 - allow empty brackets. Do not allow anything in the brackets BP5 - ignore self if included as parameter <pre> class video private name private views private starrating public procedure new(newname) name = NewName views = 0 starrating = 3 end procedure end class </pre> Examiner's Comments This question was well answered by some candidates. The question asks for pseudocode or program code and candidates should be encouraged to do one or the other if given a choice, rather than a combination of the two. Many candidates did not use the information in the question stem to help them structure their answer and gave more than one parameter in the constructor. Exemplar 3 <pre> Class Video private name private number of views private star rating public procedure new(aName) name = aName number of views = 0 star rating = 3 end procedure end class </pre> This was a good clear example of an answer given in pseudocode. The candidate has declared the 3 given attributes as private, shown a constructor with one parameter and set name to the parameter and views and rating to 0 and 3. The candidate gained 7 marks.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	- Method definition <u>that is public</u> - View attribute incremented by one	2	public procedure updateviews() views = views + 1 end procedure View attribute must have the same name as part i <u>Examiner's Comments</u> Most candidates were able to gain at least 1 mark for this question. Those who were not given marks used pseudocode but did not state that the procedure was public, or they did not use the same attribute they had declared in Question 2 (g) (i).
			Total	38	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance															
3	a	i	• 1011 0111 1110	1	CAO															
		ii	• -149	1	CAO															
		iii	0011 1001 • One mark for correct left hand nibble (CAO) • One mark for correct right hand nibble (CAO) • One mark for working clearly shown	3	Working could include showing “borrowing” values from other columns or making the second number negative and adding. Answer must be 8 bits to achieve full marks (stated in question). No marks if only working is denary Examiner’s Comments Most candidates were able to gain some marks, with many gaining full marks. A popular method was to do two’s complement addition. Candidates should be encouraged to show their working in binary and not do the subtraction in denary and then just give the answer in binary. The question asks them to complete a binary subtraction.															
	b		• Exponent is -2 • Move decimal place 2 places <u>left</u> • Fill with 1s giving 1.1101 • Denary answer is $-0.1875 // \frac{-3}{16}$	4	Accept alternative method of -0.75×2^{-2} for BP2 and BP3. BP2 credited for -0.75, BP3 credited for multiplying by 2^{-2} Correct answer with valid working gets full marks Examiner’s Comments Most candidates correctly identified the exponent and that the point needed to be moved to the left. Any valid method of working was given marks here, where candidates got the correct answer.															
	c		<table><tr><th>Binary number</th><th>Normalised</th><th>Not normalised</th></tr><tr><td>010101 100</td><td>x</td><td></td></tr><tr><td>110101 111</td><td></td><td>x</td></tr><tr><td>011010 010</td><td>x</td><td></td></tr><tr><td>101010 110</td><td>x</td><td></td></tr></table>	Binary number	Normalised	Not normalised	010101 100	x		110101 111		x	011010 010	x		101010 110	x		4	One mark per row. No mark if both/neither box is ticked. Accept other marks that clearly indicate choice (e.g. X) Examiner’s Comments This question was mostly well answered. However, some candidates were able to identify 01 as being normalised but not 10 being normalised.
Binary number	Normalised	Not normalised																		
010101 100	x																			
110101 111		x																		
011010 010	x																			
101010 110	x																			
			Total	13																

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
4	Mark Band 3–High Level (7-9 marks) The candidate demonstrates a thorough knowledge and understanding of artificial intelligence, machine learning and how biases can be inherited. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. Mark Band 2-Mid Level (4-6 marks) The candidate demonstrates reasonable knowledge and understanding of artificial intelligence, machine learning and/or how biases can be inherited; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. Mark Band 1-Low Level (1-3 marks) The candidate demonstrates a basic knowledge of some aspects of artificial intelligence or machine learning; the material is basic and contains some inaccuracies. The candidate makes a	9 (AO1.1) (2), (AO1.2) (2), (AO2.1) (2), (AO3.3). (3)	AO1 Artificial intelligence is used for computer systems that are required to perform tasks that normally require human intelligence. AI is generally programmed. Machine learning is where a machine improves its performance/output through experience/access to data. Machine learning is a subset of AI. AO2 AI follows programming so if a program contains biases then so will the AI outcomes. Programmers must be aware of this and tackle it during design/implementation/testing. Machine learning bias depends on data given. Larger data sets are generally more inclusive/less biased but must be aware of data that is itself biased. Candidates may give examples that meet this. AO3 Thorough testing, multiple programmers and scrutiny of data sets are essential to ensure that unconscious biases are not inherited. If not thoroughly considered then biases will certainly be inherited. Datasets for machine learning need to be large enough to be representative but not so large that data is not able to be checked. <u>Examiner's Comments</u> There were a wide range of responses to this question and a wide range of marks given. The most successful responses were able to address all parts of the question and could give the meaning of AI as well as some relevant examples of AI bias. They were also able to give at least one measure that could be taken. Many candidates used self-driving cars as their only example and should be encouraged to explore the use of AI in different fields. Some confused the AI being biased with people being biased against AI. Many candidates were unable to give relevant measures for preventing bias. Candidates should be encouraged to make sure they include all points the question asked for, in their answer.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			limited attempt to apply acquired knowledge and understanding to the context provided. The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 marks No attempt to answer the question or response is not worthy of credit.		
			Total	9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
5	a	i	• 40	1	CAO
		ii	• 70	1	CAO
		iii	• 300	1	CAO
	b	i	• To convert (high-level or assembly) code to <u>low level/machine code</u>	1	Do not allow answers referring to making the program executable, given in question. <u>Examiner's Comments</u> This question was generally well answered, though some candidates thought that a translator translated machine code into source code or that it translated code into something the computer could understand, without specifying what that was.
		ii	• Compiler translates code all at once/before it's executed • Interpreter translates code line by line / during runtime • Compiler produces executable file for reuse // Doesn't need to be translated everytime it is run • Interpreter needs to re-translate next time program is run • Compiler lists all errors//Compiled code doesn't run if there are any errors • Interpreter stops at the first error • Compiled programs have the source code hidden • Interpreted programs have the source code visible	4	Mark answers in pairs Max 2 marks per answer space <u>Examiner's Comments</u> This question was well answered by many candidates who showed two distinct differences.

Mark Scheme

Question			Answer/Indicative content				Marks	Guidance
	c			Lexical analysis	Syntax analysis	Code generation	5	One mark per row. No mark if more than one/no box is ticked. Accept other marks that clearly indicate choice (e.g. X)
			Comments and white-space are removed	x				
			Keywords are replaced with tokens	x				
			Object code is created			x		
			Symbol table created for variables	x				
			Builds an abstract syntax tree		x			
	d		• To make the program run faster// code is more efficient • To make the program use fewer resources/less memory				2	<u>Examiner's Comments</u> Many candidates gave good descriptions of how code is optimised, but they did not answer the question which asked what the purpose of optimisation is.
	e	i	<pre>function countCapitals(text) // initialise counter to 0 capCount = 0 // loop through each character in the string passed in for x = 0 to text.length-1 c = text.subString(x, 1) // check if character is a capital if asc(c) >= 65 and asc(c) <= 90 // if so, increment counter capCount = capCount + 1 endif next x return capCount endfunction</pre>				3	Accept alternative answers in high-level languages (e.g. capCount ++ or capCount +=1 / && for and for BP1) Accept countCapitals = capCount for BP3 (Returning via assigning to function identifier is used in VB / Pascal) Accept and asc(c) < 91 instead of <=90 Allow FT for returning the value they increment as a counter if it isn't capCount

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	- Both (use binary) to represent characters // are character sets - The first 7/8 bits of Unicode is the same as ASCII (overlaps)	1	
		iii	- ASCII has fewer characters (128/256) // Unicode has more characters - ASCII is 7/8 bits whereas Unicode can be larger 16/32 / can have variable sized characters - ASCII limited to Latin / English / European characters whereas Unicode can represent other symbols (e.g. Chinese/Cyrillic/Emojis)	2	Only mark the 1st answer for each difference For BP2&3, must have both sides to get the mark <u>Examiner's Comments</u> Candidates should be encouraged to give a complete answer. Just saying 'Unicode uses more bits than ASCII' is not enough to be given a mark at this level.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	f	Mark Band 3–High Level (7-9 marks) The candidate demonstrates a thorough knowledge and understanding of both waterfall and the spiral model. The material is generally accurate and detailed. The candidate is able to apply their knowledge and understanding directly and consistently to the context provided. Evidence/examples will be explicitly relevant to the explanation. The candidate provides a thorough discussion which is well balanced. Evaluative comments are consistently relevant and well-considered. There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated. Mark Band 2-Mid Level (4-6 marks) The candidate demonstrates reasonable knowledge and understanding of waterfall and/or the spiral model; the material is generally accurate but at times underdeveloped. The candidate is able to apply their knowledge and understanding directly to the context provided although one or two opportunities are missed. Evidence/examples are for the most part implicitly relevant to the explanation. The candidate provides a sound discussion, the majority of which is focused. Evaluative comments are for the most part appropriate, although one or two opportunities for development are missed. There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence. Mark Band 1-Low Level (1-3 marks) The candidate demonstrates a basic knowledge of some aspects of either waterfall or the spiral model; the material is basic and contains some inaccuracies. The candidate makes a limited attempt to apply acquired knowledge and	9 (AO1.1), (2) (AO1.2), (2), (AO2.1) (2), (AO3.3). (3)	AO1 The spiral model has four quadrants (determine objectives, identify and manage risk, develop and test, plan next iteration). Client feedback then informs future development and prototypes which feedback into future revisions. Waterfall has a structured analysis/design/development/test flow. Progress to the next step is not made until the previous step is completed. AO2 The spiral model relies on frequent client feedback. Spiral produces functional prototypes where features are added incrementally. Spiral model has more focus on risk; projects may be modified or even dropped if risk is too great. Waterfall is much more structured and very reliant on getting the definition of requirements correct at the start; changes are harder to add in at a later stage. However, this forces the definition to be well understood. AO3 Spiral involves client feedback, prototypes and evolving projects. Better option where requirements may change. Waterfall is better where requirements are very clear to begin with and outcomes known. Spiral is better for risk management. If the programmer has a large team then waterfall may be more appropriate due to the clearly defined responsibilities at each stage. Examiner's Comments Most candidates gained some marks on this question. Most could explain that spiral was iterative and waterfall was done in linear stages, but many did not expand on this. Few candidates linked their answer to the complex computer program mentioned in the question. Some candidates also described waterfall as iterative.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			understanding to the context provided. The candidate provides a limited discussion which is narrow in focus. Judgments if made are weak and unsubstantiated. The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear. 0 marks No attempt to answer the question or response is not worthy of credit.		
			Total	30	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6	a	i	- For printer queue - All documents/users have equal priority - Whichever document is received first is printed first - First in First Out / Last in Last Out	2	Examiner's Comments This question was generally well answered with many candidates gaining at least 1 mark. The question asked why the OS used 'first come first served' and many candidates were not given marks for stating why the user would want the OS to use it. Some candidates talked about other scheduling algorithms even though this was not relevant to the question.
		ii	- To enable <u>multitasking</u> to take place - To switch between active processes and those running in the background - To limit each process to a certain amount of time//allow processes an equal share of processor time.... ...to ensure the OS cycles through all processes // the process then goes to the back of the end of the queue ...so that users can receive an immediate response ...to handle an interrupt immediately	3	BP4, 5 & 6 are dependent on BP3 only
		iii	- Shortest job first / shortest remaining time - Process which has the shortest <u>time</u> (remaining) is completed first - Multilevel feedback queues - Uses <u>queues</u> with different priorities - Jobs can be moved between <u>queues</u>	2	One mark for name, one mark for description.
	b	i	- Protocol to be used is decided based on the application - E.g. HTTPS for browser based service // SMTP/IMAP for messaging service - Adds encryption - Passes on <u>to</u> transport layer to send - Gets data <u>from</u> transport layer when receiving - Unpacks message ready for display // removes headers or other non-viewable data - Decrypts message	5	For BP2, don't allow HTTP (question mentions encryption). Don't allow a list of protocols which aren't relevant to the question. Don't allow a protocol without its use Examiner's Comments Very few students could explain what happens at the application layer and answers tended to be about splitting data into packets. Some candidates did mention that encryption would take place but didn't go on to mention decryption when receiving data. Those candidates that identified that protocols are applied here were unable to give a specific example and simply listed protocols they knew, but without context.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	- Receives (layered) data <u>from</u> internet layer to send - MAC addresses are added to the packet - Passes and receives data across wireless network (to WAN / other machine) - Passes (layered) data back up <u>to</u> internet layer when receiving	2	Wireless access given in question stem <u>Examiner's Comments</u> Very few candidates were able to gain 2 marks on this question. Some candidates talked about transmitting data via cables, despite wireless being mentioned in the question.
			Total	14	